

What is Statistics?

Definition

Statistics is the science of

- Collecting data
- Organizing data
- Summarizing data
- Analyzing data
- Interpreting data
- Making decisions using data

Simply,

Statistics is the science of learning from data.

Real-life Examples

Hospital

Questions doctors ask:

- Does this medicine work?
- What is the recovery rate?

- Which treatment is better?

Statistics provide the answers.

Why Do We Need Statistics?

Without statistics,

Imagine a company has sales data for 5 million customers.

Can a manager understand millions of rows manually?

No.

Statistics summarizes the data into meaningful insights.

Statistics helps us:

- Understand data
- Find trends
- Compare groups
- Make predictions
- Support business decisions

Types of Statistics

Statistics has two major branches.

None

Statistics

|

├— Descriptive Statistics

|

└— Inferential Statistics

Descriptive Statistics

Definition

Descriptive statistics summarizes the available data.

It tells us

"What happened?"

It **does not** predict.

Example

Marks

None

70

80

65

90

75

Instead of looking at all numbers, we calculate

- Average
- Highest
- Lowest
- Median
- Standard deviation

Common Measures

- Mean
- Median

- Mode
- Range
- Variance
- Standard Deviation
- Quartiles
- Percentiles

Visualizations

- Bar Chart
- Histogram
- Pie Chart
- Box Plot
- Line Chart

Real-Life Example

Monthly sales

None

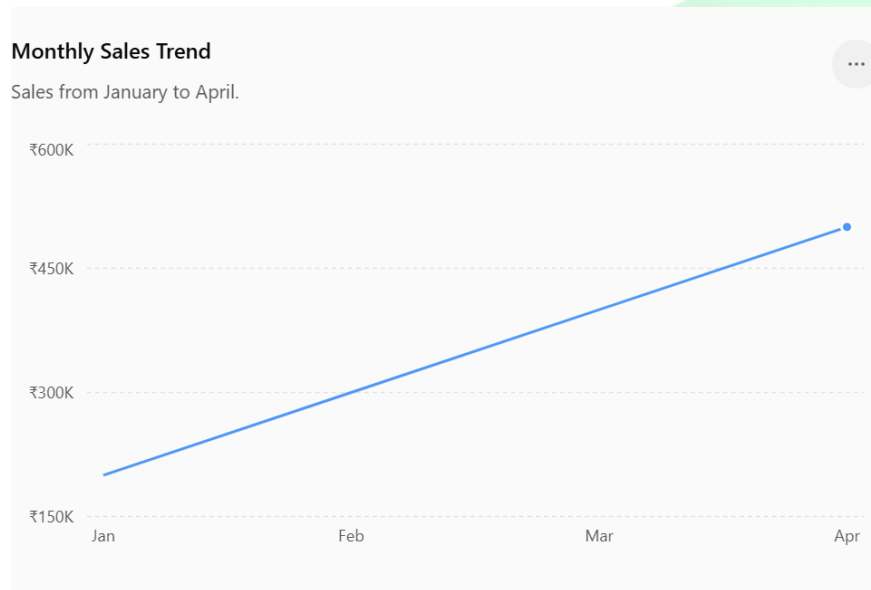
Jan 2 Lakh

Feb 3 Lakh

Mar 4 Lakh

Apr 5 Lakh

Instead of reading numbers, we create a line chart.



That is descriptive statistics.

Advantages

- Easy to understand
- Summarizes large datasets
- Supports quick decisions

Limitation

Cannot predict the future.

Inferential Statistics

Suppose India has

1.4 billion people.

Can we ask everyone

"What mobile phone do you use?"

Impossible.

Instead,

Survey

10,000 people.

Predict the result for the whole country.

This is Inferential Statistics.

Definition

Inferential statistics uses sample data to make conclusions about the entire population.

Example

Population

50,000 students

Sample

500 students

Average CGPA

8.2

Inference

The average CGPA of the college is approximately 8.2.

Techniques

- Sampling
- Confidence Interval
- Hypothesis Testing
- Regression
- ANOVA

- Correlation

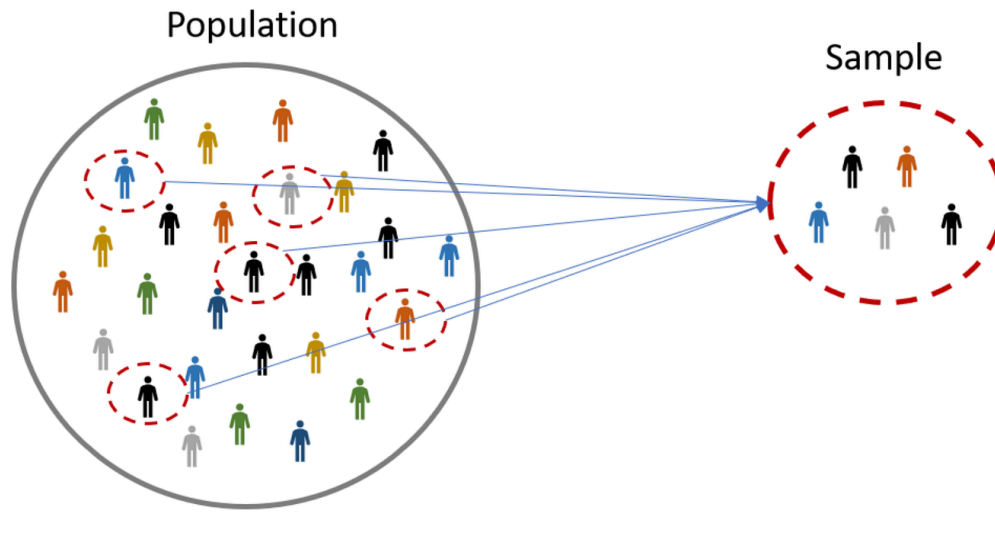
Applications

- Election Poll
- Medical Research
- Customer Survey
- Weather Forecast
- Stock Market

Difference

Descriptive	Inferential
Summarizes data	Predicts population
Uses available data	Uses sample
No prediction	Prediction
Charts	Probability

Population and Sample



Population

- Entire collection.
- Examples
- All Indians
- All Amazon users
- All students in GUVI
- All customers of Flipkart

Represented by **N**

Sample

Small subset selected from population.

Represented by

n

Example

Population

- 10,000 employees

Sample

- 500 employees

Why Sampling?

Suppose WHO wants to test a vaccine.

Testing all humans?

Impossible.

Instead,



Take

5,000 volunteers.

That is sampling.

Parameter vs Statistic

Parameter	Statistic
Population	Sample
μ	$\bar{x}$
σ	s

Types of Sampling

There are **two major categories**:

1. **Probability Sampling**
2. **Non-Probability Sampling**

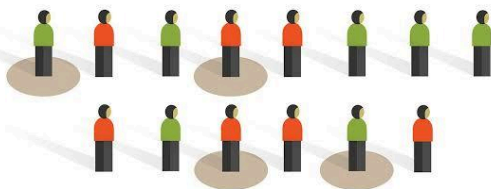
1. Probability Sampling

Every individual has a **known and equal probability** of being selected.

Suitable for statistical analysis because it reduces bias.

A. Simple Random Sampling

Simple random sampling



Every member has an equal chance of being selected.

Example

Population:

None

100 students

Need a sample of 10 students.

Randomly select using:

- Random number generator
- Lottery method

Example Selected IDs:

None

5, 12, 24, 30, 45, 51, 67, 78, 82, 96

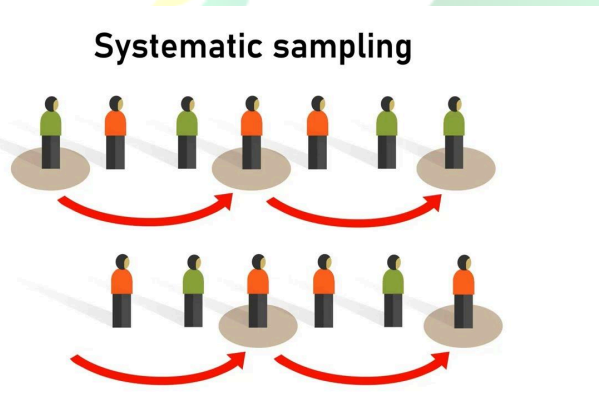
Advantages

- Easy
- Unbiased
- Fair selection

Disadvantages

- Needs complete population list
- Difficult for very large populations

B. Systematic Sampling



Select every **kth** individual.

Formula

None

$$k = \text{Population Size} / \text{Sample Size}$$

Example

Population = 1000

Need Sample = 100

None

$$k = 1000/100 = 10$$

Choose a random starting point (say 4)

Then choose:

None

4, 14, 24, 34, 44 . . .

Advantages

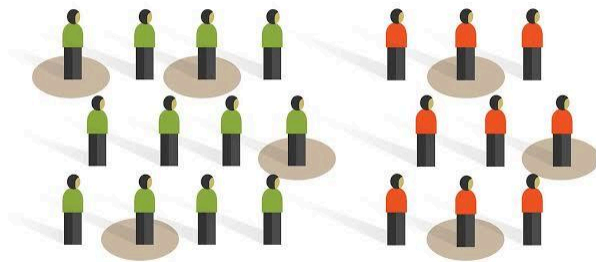
- Easy
- Fast

Disadvantages

If data has a hidden pattern, the sample may become biased.

C. Stratified Sampling

Stratified sampling



Population is divided into **homogeneous groups (strata)**.

Then random sampling is performed from each group.

Example

College students

None

Male : 600

Female : 400

Need 100 students

Take

None

Male = 60

Female = 40

Randomly select within each group.

Why use it?

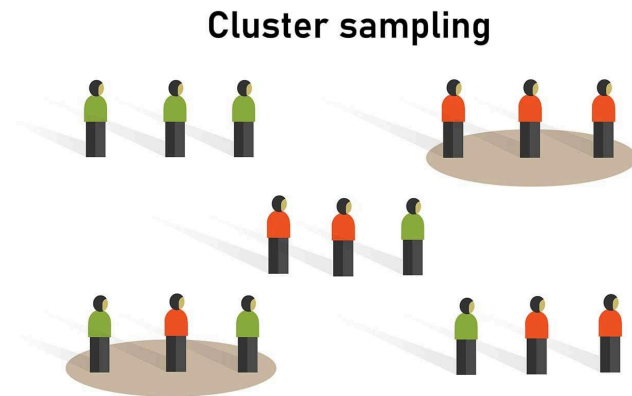


Ensures every subgroup is represented.

Applications

- Surveys
- Elections
- Medical studies

D. Cluster Sampling



Population is divided into **clusters**.

Instead of selecting individuals, entire clusters are selected randomly.

Example

Suppose India has

None

100 schools

Randomly choose

None

10 schools

Survey all students in those schools.

Difference

Stratified:

- Sample from every group

Cluster:

- Select a few entire groups

2. Non-Probability Sampling

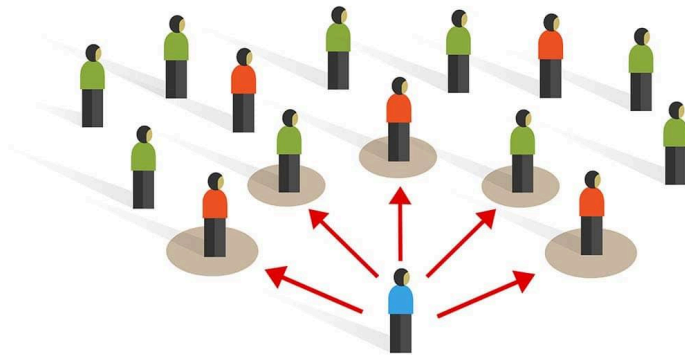
Selection is **not random**.

Some members have a higher chance of being selected.

Generally used when quick or exploratory insights are needed.

A. Convenience Sampling

Convenience sampling



Choose whoever is easiest to reach.

Example

Survey students standing near the college entrance.

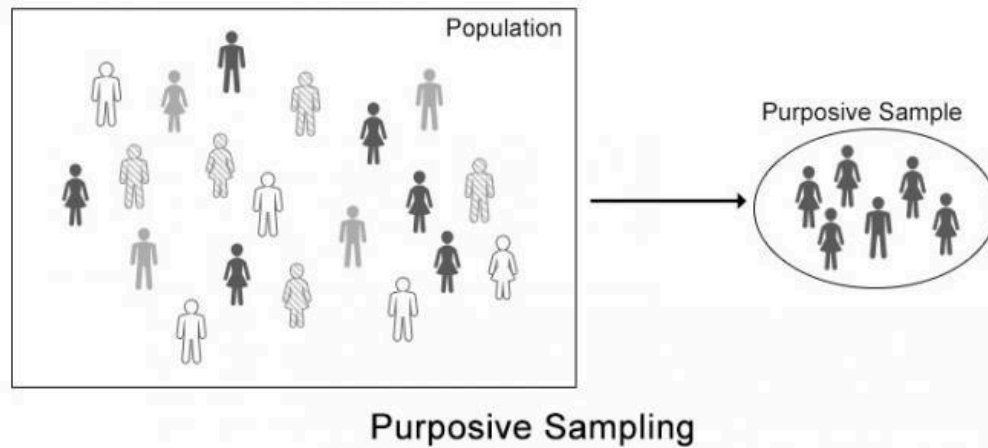
Advantages

- Fast
- Cheap

Disadvantages

Highly biased.

B. Judgment (Purposive) Sampling



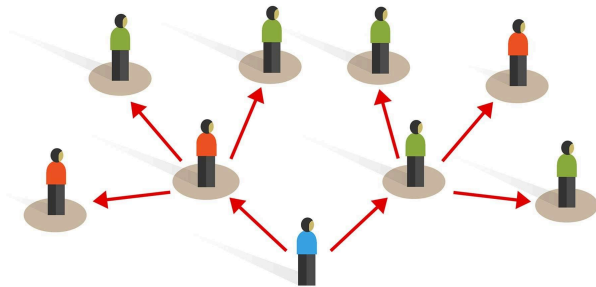
Researcher selects people based on expertise or specific characteristics.

Example

Interview only experienced doctors about a new medical policy.

C. Snowball Sampling

Snowball sampling



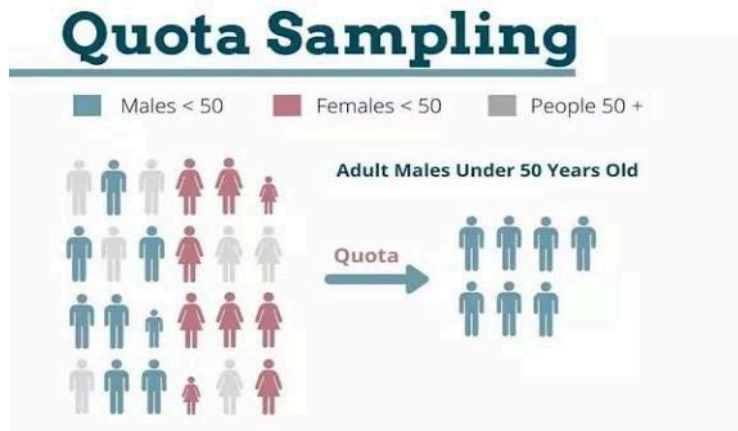
Current participants refer new participants.

Useful for hidden or hard-to-reach populations.

Example

Research on drug users or rare disease patients.

D. Quota Sampling



Population is divided into groups, and a fixed quota is filled without random selection.

Example

Need:

- 50 males
- 50 females

Select the first available participants until each quota is reached.